Prof. Dr. Benedikt Wiestler

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AI-IDT
AI for Image-Guided Diagnosis and Therapy

Summary

I am a neuroradiologist and computer science researcher specializing in the computational analysis and modeling of medical data. My work is motivated by translating these advances into practical solutions that enhance clinical decision-making and, ultimately, improve patient care.

Career

Since 05/2024	Associate W3 Professor, TUM School of Medicine & Health
09/2020 - 04/2024	Attending Physician, Department of Neuroradiology, TUM
01/2018 - 04/2024	Group Leader *Computational Imaging*, TUM
05/2015 - 06/2020	Resident, Department of Neuroradiology, TUM
07/2012 - 12/2014	Postdoc, German Cancer Research Center, Heidelberg
01/2010 - 06/2012	Resident, Department of Neuro-Oncology, Heidelberg University Hospital

Education

2019 - 2020	Certificate of Advanced Studies *"Artificial Intelligence in Medical Imaging"*, University of Bern
06/2018	Habilitation in Radiology, TUM
2015 - 2019	DKTK School of Oncology Fellow, TUM
06/2011	Doctoral Thesis (*summa cum laude*), Heidelberg University
2002 - 2009	Medical School, Bonn & Heidelberg Universities

Honors & Awards

2020	Best project, *"Image Quality Transfer in MRI with Deep Neural Networks"*, Medical Image Computing Summer School, University College London
2019	Kurt Decker Award, German Society of Neuroradiology
2018	2nd Place "Lecturer of the Year", Faculty of Medicine, TUM
2005 – 2009	Fellowship of the German National Merit Foundation (Studienstiftung)

Third-Party Funding

My research is supported by significant funding through competitive national grants (e.g., DFG individual grants and Priority Programme "Radiomics"), national cancer charities (Deutsche Krebshilfe), and federal ministries (BMBF, BMWi), as well as international grants (NIH, EU-COST).

Academic Service

I actively contribute to both national and international efforts in my field. This includes participating in groups like the RANO-AI Working Group and the "Multiple Sclerosis" guideline committee of the German Neurological Society. In addition, I serve on the review boards of multiple academic journals, including *Medical Image Analysis*, *Neuro-Oncology*, and *Radiology*, and regularly review for international conferences, such as the *Medical Image Computing and Computer Assisted Intervention* (MICCAI) conference.

Research Statement

A fundamental challenge in modern medicine is the gap between the growing availability of complex, multimodal data and its limited use in daily clinical decision-making. This disconnect between data richness and clinical utilization represents a critical barrier to advancing precision medicine. My *AI for Image-Guided Diagnosis and Therapy* lab aims to bridge this gap: We develop **state-of-the-art Artificial Intelligence models that synthesize this complex information into actionable biomarkers**, directly informing precision diagnostics and personalized therapeutic strategies.

This translational focus is reflected throughout our publications. Foundational work, such as [1] and [2], showcases collaboration with computer science groups to pioneer novel image analysis techniques and biophysical models. The clinical impact of these methods is demonstrated in studies such as [3], where we applied computational analysis to enhance the assessment of inflammatory disease activity in Multiple Sclerosis, and [4], which leverages Deep Learning models to inform personalized radiotherapy planning.

Beyond my core research, I am committed to advancing the field through community leadership, evidenced by my role as a co-organizer for internationally recognized challenges and workshops, including the Brain Tumor Segmentation (BraTS) and Ischemic Stroke Lesion Segmentation (ISLES) challenges [5].

1. McGinnis J, Shit S, Hoelzl A, Friedrich P, Bueschl P, Sideri-Lampretsa V, Muehlau M, Cattin PC, Menze BH, Rueckert D, **Wiestler B**.
   Beyond Uniformity: Regularizing Implicit Neural Representations through a Lipschitz Lens.
   ICLR 2026

2. Balcerak M, Amiranashvili T, Wagner A, Weidner J, Karnakov P, Paetzold JC, Ezhov I, Koumoutsakos P, **Wiestler B***, Menze BH* (equal contribution).
   Physics-Regularized Multi-Modal Image Assimilation for Brain Tumor Localization.
   NeurIPS 2024

3. Eichinger P, Schoen S, Pongratz V, Wiestler H, Zhang H, Bussas M, Hoshi MM, Kirschke J, Berthele A, Zimmer C, Hemmer B, Muehlau M, **Wiestler B**.
   Accuracy of Unenhanced MRI in the Detection of New Brain Lesions in Multiple Sclerosis.
   Radiology 2019

4. Weidner J, Ezhov I, Balcerak M, Metz M, Litvinov S, Kaltenbach S, Feiner L, Lux L, Kofler F, Lipkova J, Latz J, Rueckert D, Menze B, **Wiestler B**.
   A Learnable Prior Improves Inverse Tumor Growth Modeling.
   IEEE TMI 2024

5. de la Rosa E, Reyes M, Liew SL, ..., Menze BH, Kirschke J, **Wiestler B**.
   DeepISLES: A clinically validated ischemic stroke segmentation model from the ISLES'22 challenge.
   Nature Communications 2025